Sanjivani Rural Education Society's
Sanjivani College of Engineering, Kopargaon-423 603
(An Autonomous Institute, Affiliated to Savitribai Phule Pune University, Pune)
NAAC 'A' Grade Accredited, ISO 9001:2015 Certified

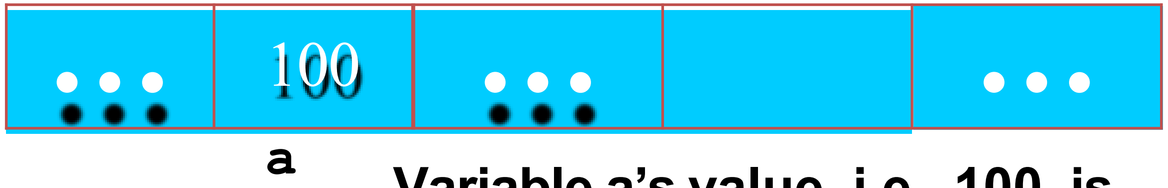
Department of Computer Engineering
(NBA Accredited)

Subject- Object Oriented Programming (CO212)
Unit 3 – Polymorphism and Virtual Function
Topic – 3.4 Pointer, Array of Pointer

Computer Memory

- Each variable is assigned a memory slot (the size of variable depends on the data type) and the variable's data is stored there in that memory slot.
- Eg: int a=100;

Memory address: 1024



Variable a's value, i.e., 100, is stored at memory location 1024

Computer Memory

- Every byte in computer memory has an address.
- Addresses are numbers just like house numbers.
- The moment a variable is declared, memory address is allocated in the RAM for its storage. int X=100;.
- A variable in the RAM can be accessed only if you know its address.

Pointer

- A pointer is a variable which stores the memory address of another variable.
- It supports dynamic memory allocation.
- **Declaration and Initialization of Pointer Variable:**

Syntax :

datatype *variable_name;

e.g. int *x;

float *y;

char *z;

- Address of operator(&)- It is a unary operator that returns the memory address of its operand.

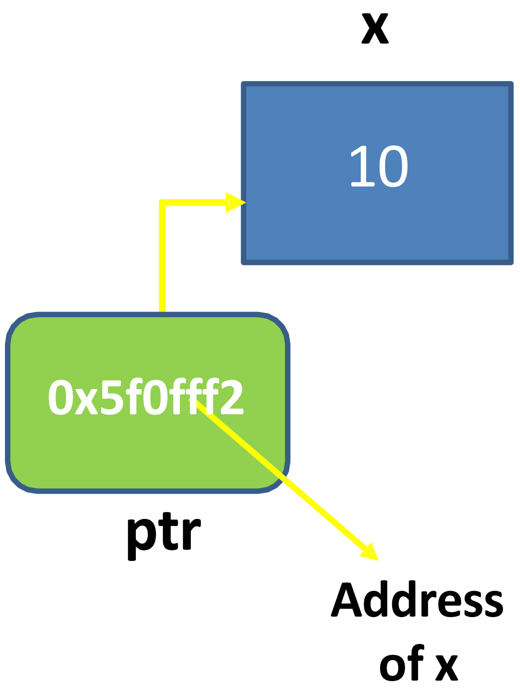
ptr_name = &variable_name;

Pointer Example

- eg. **int x = 10;**

  int *ptr ;

  ptr = &x;
- Now ptr will contain address where the variable x is stored in memory.

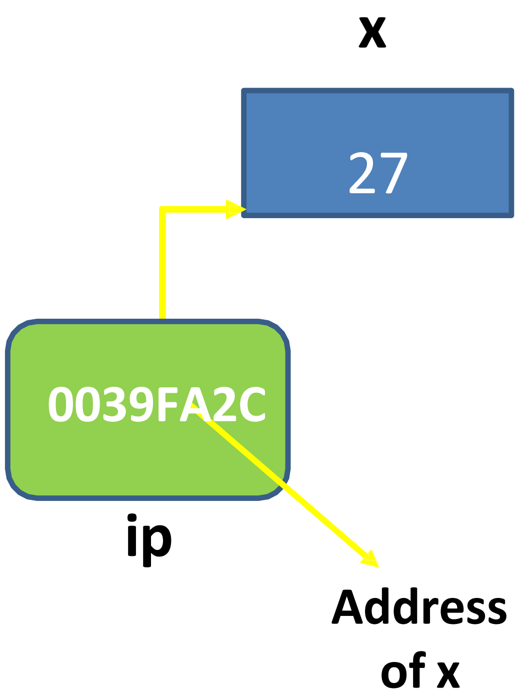


Pointer Example

```
#include <iostream>
int main()
 {
int x = 27;
int *ip;
 ip = &x;
cout << "Value of x is : "<< x << endl;
cout << "Value of ip is : " << ip << endl;;
cout << "Value of *ip is : " << *ip << endl;;
return 0;
 }
```

Output:

```
Value of x is : 27
Value of ip is : 0039FA2C
Value of *ip is : 27
```



Array of Pointer

- An array is a collection of elements of similar data type.
- We can create pointer variable with array.
- Using array of pointer searching,insertion, deletion etc. operations can be performed.
- Syntax:

```
data_type   variable_name[size];
data_type    *pointer_variable;
pointer_variable=variable_name;
OR
 pointer_variable=&variable_name[0];
```

Example1:Array of Pointer

```cpp
#include <iostream>
using namespace std;
int main()
{
    int *ptr;
    int marks[10];
    cout << "Enter the elements of an array :";
    for(int i=0;i<10;i++)
    {
        cin>>marks[i];
    }
    ptr = marks;            //marks and ptr pointing to same element
    cout << "The value of *ptr is :" <<*ptr;
    cout << "The value of *marks is :" <<*marks;
}
```

```
Enter the elements of an array :
1
2
3
4
5
6
7
8
9
10
The value of *ptr is :1
The value of *marks is :1
```

Example2:Array of Pointer

```
int main()
{
     int a[5];
     int *ptr2[5];
    cout << "Enter five numbers :“;
    for(int i=0;i<5;i++)
    {
        cin >> a[i];
    }
    for(int i=0;i<5;i++)
    {
        ptr2[i]=&a[i];
    }
    cout << "The values are”;
    for(int i=0;i<5;i++)
    {
       cout << *ptr2[i] ;
    }
}
```

```
Enter five numbers :
4
5
1
2
3
The values are
4
5
1
2
3
```

Thank You..